FORK. BUILD. MERGE. · ROAD TO DEVFEST 2026

# Namma Commit

## + MergeWarden

First-timers guided to a merged PR on WhatsApp, in their own language, and every PR reviewed against the repo’s own rules.

**WhatsApp coach · any language** | Google ADK multi-agent | Gemini

github.com/gdgcblr/open-source-agents/pull/5 · agents/mergewarden

CHAPTER 1 · MEET ANANYA

# Ananya wants to contribute to open source. She has four questions.

**1**

### Which repo?

Millions of projects. Which are active, friendly, and right for my skills?

**2**

### What do I contribute?

Issues are vague. What is a small, real first task I can actually finish?

**3**

### How do I contribute?

Fork, branch, PR, the repo's rules: all jargon, and all in English.

**4**

### Will it get merged?

The PR waits in a queue, then bounces on a rule they never saw.

Most students like Ananya give up at question 1, 2 or 3. The ones who reach 4 land in a maintainer's overflowing queue.

Ananya is an illustrative second-year engineering student from Mysuru · her questions are the real ones

CHAPTER 2 · ANANYA ISN’T ALONE

# It’s happening right now, in this event’s repo

**166**
PRs in 2 h 26 min

**82**
of them in a single hour

**~2,150**
lines added in the median PR

**0**
merged so far

### 154 people, 233 forks

Every one of them is making a first contribution to this repo, today.

### The rules got missed

22% of PR titles ignore the repo’s convention: “pre changes for the tool”, “Feat/merge ready”.

### One queue for everyone

Every PR still waits for a human to check the same 8-item list by hand.

Source: GitHub API, gdgcblr/open-source-agents, 2026-10-24 06:49–09:15 UTC

CHAPTER 3 · WHAT IF

# What if one WhatsApp chat could answer all four of Ananya’s questions?



## Namma Commit

The student’s coach on WhatsApp · questions 1–3

- Finds the right repo and first issue for you
- Explains it and plans the fix with you
- Guides fork → PR, in any language, by voice or text



## MergeWarden

The maintainer’s reviewer · question 4

- Checks each PR against the repo’s own rules
- Cites the rule and the fix for every finding
- Ranks the whole queue, ready PRs first

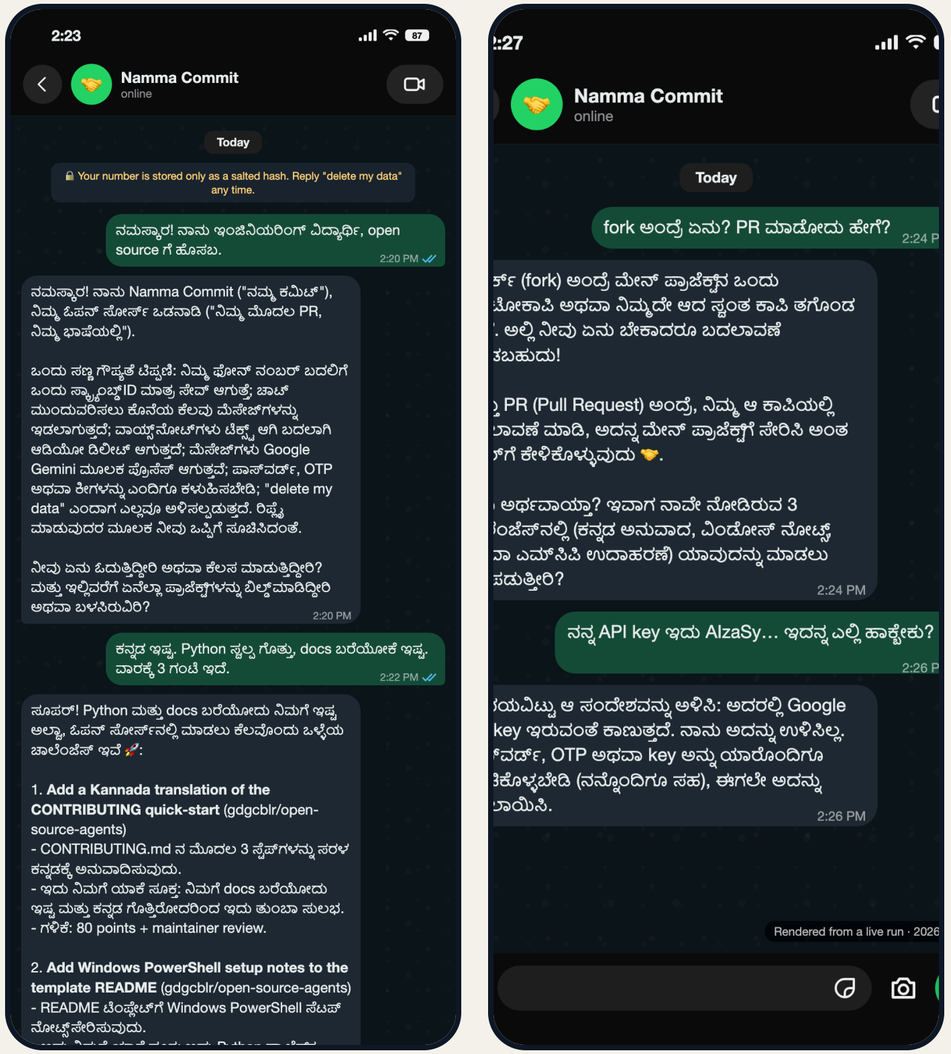
The hand-off: the student says “check my PR” → MergeWarden reviews it → the coach explains the result in the student’s language → they fix it before a maintainer ever looks.

CHAPTER 4 · ANANYA SAYS “ನಮಸ್ಕಾರ”

# Namma Commit answers her first three questions, in Kannada

1. **Which repo?** With consent, it reads your public GitHub and picks active, welcoming repos that match your skills, each with a “why it fits”.
2. **What to contribute?** Real good-first issues or event challenges, explained with the repo’s own CONTRIBUTING.md, plus a FIX_PLAN with a ready-to-paste Copilot prompt.
3. **How?** Fork → branch → PR, one step per message, with the exact commands for your OS. It asks you questions back instead of just doing it for you.

Safe by design: keys are refused before any model call, phone numbers are hashed, it never runs code, and points come only after GitHub verifies the PR. Tested end to end in the simulator. The Twilio adapter is ready, and the real-phone round trip is pending.




Screenshots: real replies from a live Gemini run in Kannada, shown in a WhatsApp-style frame

CHAPTER 5 · SEVEN MESSAGES LATER

# Ananya goes from “hi” to a reviewed PR

**01 · say hi**

“ನಮಸ್ಕಾರ!” Privacy note, then three questions: language, skills, time.

**02 · get matched**

Reads public GitHub with consent. Three picks, each with a “why it fits”.

**03 · understand it**

Issue deep-dive with the repo’s CONTRIBUTING.md, plus questions back.

**04 · plan the fix**

Talk it through, even as a voice note. Get FIX_PLAN.md and a Copilot prompt. +10 points.

**05 · open the PR**

Fork → branch → commit → PR, one step per message, exact commands.

**06 · “check my PR”**

Authorship verified on GitHub. MergeWarden reviews it in the background.

**07 · fix and repeat**

Verdict and top fixes, in Kannada. Push, then “review again”.

✅ ready

**+20 for the PR, +50 when merged, and the Clean PR badge.**

Points only after the GitHub API confirms the student authored the PR · 5-minute cooldown, 5 reviews a day

CHAPTER 6 · BEHIND THE SCENES

# Her PR meets MergeWarden: rules first, AI judgement second




### Rules from the base branch

Every finding cites its line in CONTRIBUTING.md. A PR can’t weaken its own rules.

### AI must quote the PR

A finding whose quote isn’t in the contributor’s files is demoted to a note.

### AI can’t gate

It can’t block, clear a rule, approve, post or merge. A human decides.

CHAPTER 7 · KEEPING ANANYA SAFE

# She’s a student, so the guardrails come first

### Secrets refused

A pasted key or OTP is refused before any model call, never stored, and the student is told to rotate it.

### Privacy

Phone numbers stored only as salted hashes. Voice bytes dropped after transcription. “delete my data” wipes it all.

### Untrusted input

Messages and PR text are wrapped as data, injection attempts are flagged, and the AI can’t act on them.

### No code execution

PRs are snapshotted, never run on the host. Coach drafts stay in a per-student sandbox.

### Fair points

Only after GitHub verifies authorship. Append-only ledger, daily cap, nothing farmable.

### Human decides

Nothing is posted, pushed or merged automatically. Signed webhooks, rate limits, localhost by default.

CHAPTER 8 · HOW IT'S BUILT

# Google ADK agents talk to Ananya, and code makes the decisions

### `coach_agent · LlmAgent coordinator`

Sub-agents: onboarding · issue scout · mentor · solution drafter (FixPlan schema) · PR guide

Deterministic tools: analyze_github_profile · find_matching_work · verify_pr · award_points · save_fix_plan · run_preflight_checks · review_my_pr

Served by FastAPI: WhatsApp (Twilio, signed webhook), web simulator, terminal

### `maintainer_copilot · LlmAgent coordinator`

Tools: review_contribution · triage_open_prs · suggest_fixes, with coach_agent as a sub-agent

Review pipeline (SequentialAgent): evidence_collector → ParallelAgent(doc_reviewer, policy_reviewer) → arbiter → review_writer → renderer

Interfaces: CLI · web console · adk web · exit codes 0/1/2 as a CI gate

**Models.** Reviews: Gemini 3.5 Flash Lite → 3.8 Flash. Coach: 3.8 Flash → 3.5 Flash Lite. Both fall back to deterministic-only.

**Offline first.** With no API key, every rule still runs. Gemini adds judgement, and never the gate.

CHAPTER 9 · DOES IT WORK?

# We measured it on this event’s own pull requests

**44 s**
to triage all 5 open upstream PRs

**~9 s**
per review with Gemini

**100**
its own PR’s score: Ready, 100/100

**18/18**
labelled eval cases correct

**138**
automated tests pass, offline

**0**
comments posted on other teams’ PRs

It flagged a real PR whose ticked checklist was contradicted by the evidence. Gemini caught an unbounded while True API loop that no rule covers.

All figures from runs on 2026-10-24 · gdgcblr/open-source-agents · evals/RESULTS.md

CHAPTER 10 · WHY THIS ONE

# Most tools meet Ananya after her PR. We meet her before it exists.

### Before the PR

Helps a student who has no PR yet: the repo, the issue, the plan.

### Your language

Any language, text or voice, on WhatsApp, where students already are.

### The repo’s own rules

Compiled from CONTRIBUTING.md on the base branch, with line citations.

### AI never gates

Findings must quote the PR. The AI can’t block, approve, post or merge.

### Measured

18/18 evals, 138 tests, and triage of this repo’s real PRs.

### Honest about limits

We red-teamed it ourselves and published the gaps we’re closing.

22 of the 166 PRs here on 2026-10-24 are PR-review, preflight or first-PR tools. The need is real.

EPILOGUE

# Ananya’s first commit becomes namma commit, “our commit”

```
cd agents/mergewarden && pip install -r requirements.txt
python -m mergewarden.coach serve # the coach
python -m mergewarden review <PR> --no-llm
python -m mergewarden triage <owner/repo>
```

## Next

- Real-phone WhatsApp round trip
- Close the red-team gaps we found
- GitHub Action for every new PR
- Voice replies and a mentor dashboard

Every student’s first commit should feel like theirs. Every maintainer’s queue should feel manageable.